Interop QOS

2026-03-10-15-24-36

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions.By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels,ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

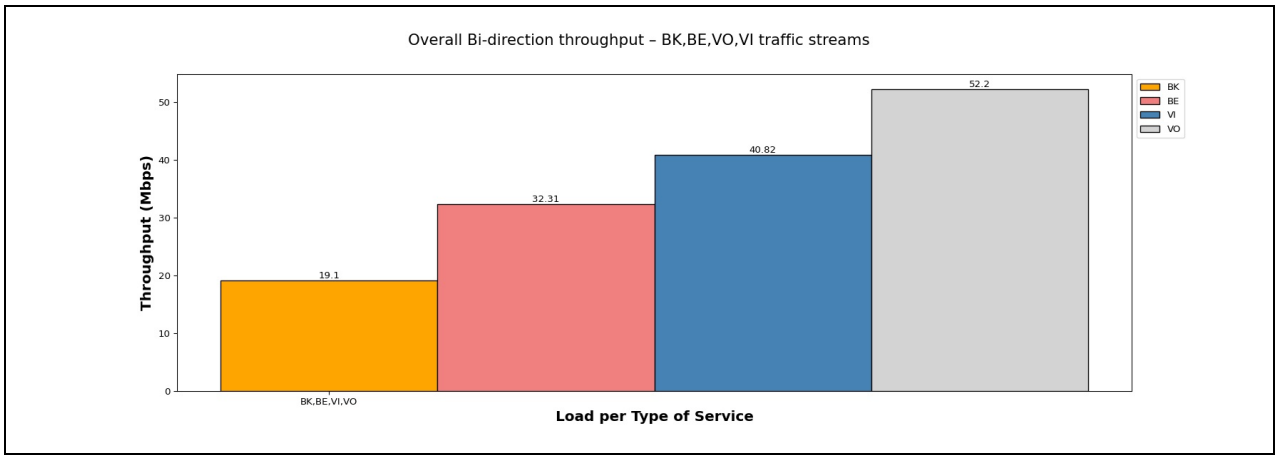
Test Configuration	Device List	DESKTOP-DDPI3HE(Windows), vivof(Android), ubuntu24(Linux)
	Number of Stations	Total(3) Android(1) Windows(1) Linux(1)
	AP Model	NETGEAR
	SSID	NETGEAR_2G_wpa2
	Traffic Duration in hours	0.02
	Security	wpa2
	Protocol	UDP
	Traffic Direction	Bi-direction
	TOS	['VI', 'VO', 'BE', 'BK']
	Per TOS Load in Mbps	Upload:100.0 Mbps,Download:100.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 100.0 Mbps-upload	Throughput for Load 100.0 Mbps-download
3	BK : 11.12, BE : 15.06, VI: 26.85, VO: 34.26	BK : 7.98, BE : 17.25, VI: 13.97, VO: 17.94

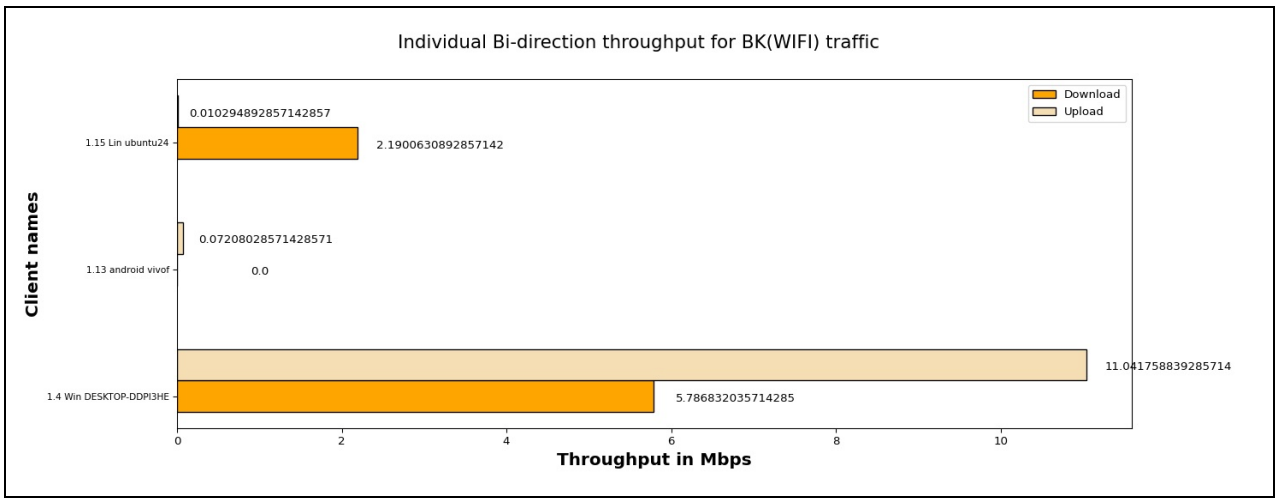
Overall Bi-direction throughput for 3 clients with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loadsUpload:100.0 Mbps,Download:100.0 Mbps per tos



Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BK(WiFi).

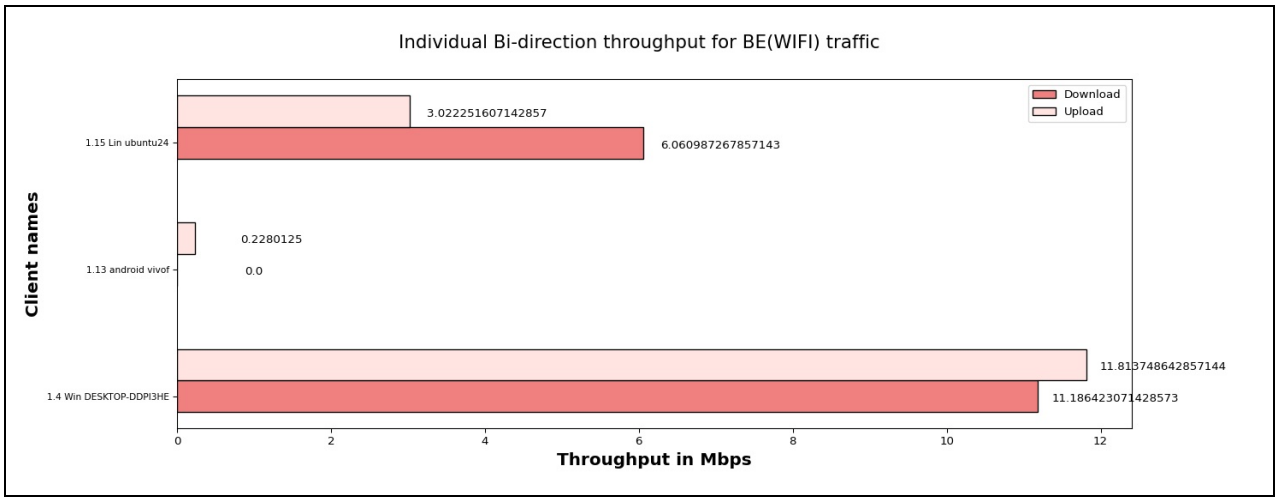
The below graph represents individual throughput for 3 clients running BK (WiFi) traffic. X- axis shows “Throughput in Mbps” and Y-axis shows “number of clients”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.4 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_5G_wpa2	Background	100.0 Mbps	100.0 Mbps	11.041758839285714 Mbps	5.786832035714285 Mbps	0.000000	61.006113	30	FAIL
1.13 android vivof	6:97:12:01:52:0	NETGEAR_2G_Open	Background	100.0 Mbps	100.0 Mbps	0.07208028571428571 Mbps	0.0 Mbps	0.440529	62.498544	30	FAIL
1.15 Lin ubuntu24	c:5f:67:69:3c:86	NETGEAR_2G_wpa2	Background	100.0 Mbps	100.0 Mbps	0.010294892857142857 Mbps	2.1900630892857142 Mbps	80.827664	73.983008	30	FAIL

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BE(WiFi).

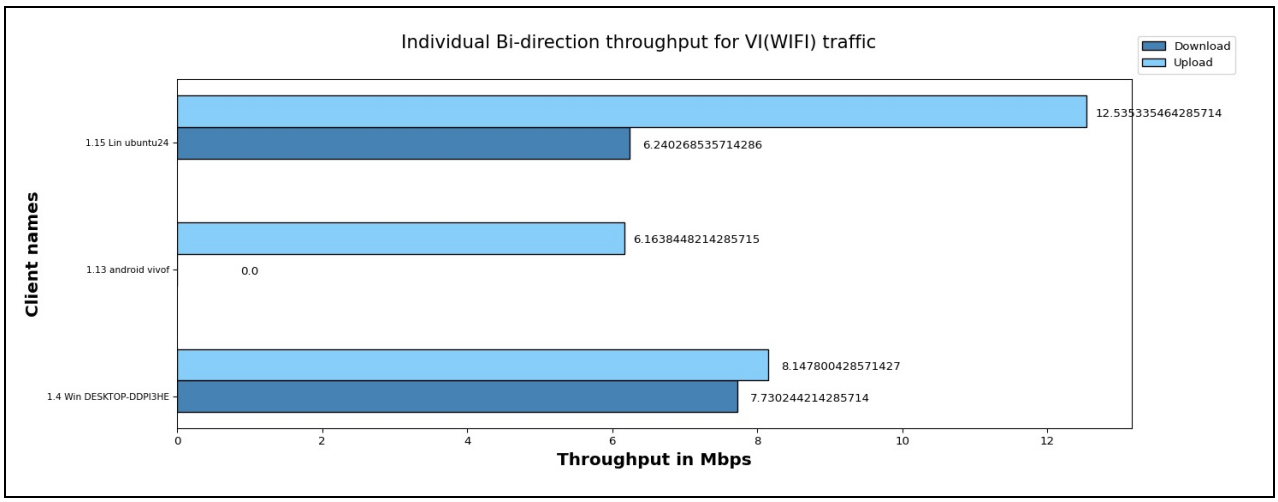
The below graph represents individual throughput for 3 clients running BE (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.4 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_5G_wpa2	Besteffort	100.0 Mbps	100.0 Mbps	11.813748642857144 Mbps	11.186423071428573 Mbps	0.000000	50.771644	30	FAIL
1.13 android vivof	6:97:12:01:52:0	NETGEAR_2G_Open	Besteffort	100.0 Mbps	100.0 Mbps	0.2280125 Mbps	0.0 Mbps	0.000000	71.428571	30	FAIL
1.15 Lin ubuntu24	c:5f:67:69:3c:86	NETGEAR_2G_wpa2	Besteffort	100.0 Mbps	100.0 Mbps	3.022251607142857 Mbps	6.060987267857143 Mbps	55.740892	72.973563	30	FAIL

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VI(WiFi).

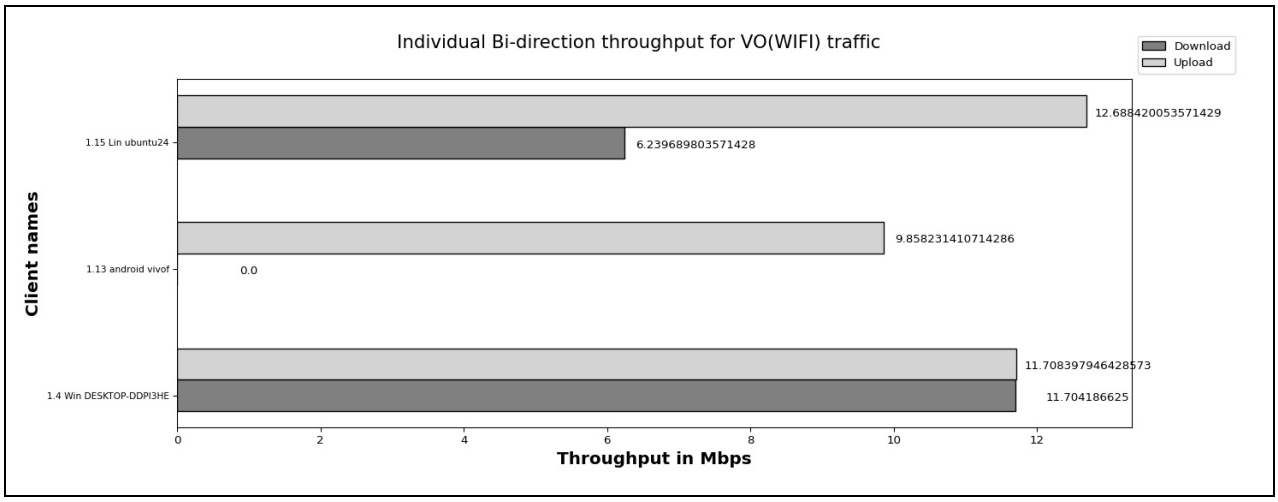
The below graph represents individual throughput for 3 clients running VI (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.4 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_5G_wpa2	Video	100.0 Mbps	100.0 Mbps	8.147800428571427 Mbps	7.730244214285714 Mbps	0.000000	35.970832	30	FAIL
1.13 android vivof	6:97:12:01:52:0	NETGEAR_2G_Open	Video	100.0 Mbps	100.0 Mbps	6.1638448214285715 Mbps	0.0 Mbps	0.000000	37.416352	30	FAIL
1.15 Lin ubuntu24	c:5f:67:69:3c:86	NETGEAR_2G_wpa2	Video	100.0 Mbps	100.0 Mbps	12.535335464285714 Mbps	6.240268535714286 Mbps	56.397453	62.470749	30	FAIL

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 3 clients running VO (WiFi) traffic. X- axis shows “number of clients” and Y-axis shows “Throughput in Mbps”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)	Expected Bi-direction rate(Mbps)	Status
1.4 Win DESKTOP-DDPI3HE	e4:a4:71:93:70:e7	NETGEAR_5G_wpa2	Voice	100.0 Mbps	100.0 Mbps	11.708397946428573 Mbps	11.704186625 Mbps	0.000184	48.107677	30	FAIL
1.13 android vivof	6:97:12:01:52:0	NETGEAR_2G_Open	Voice	100.0 Mbps	100.0 Mbps	9.858231410714286 Mbps	0.0 Mbps	0.000000	80.319591	30	FAIL
1.15 Lin ubuntu24	c:5f:67:69:3c:86	NETGEAR_2G_wpa2	Voice	100.0 Mbps	100.0 Mbps	12.688420053571429 Mbps	6.239689803571428 Mbps	68.259763	62.933576	30	FAIL

Information	contact	support@candelatech.com